# ATM: Adaptive Arbitrary-Length Tokenization with Semantic Modeling for Time Series Forecasting

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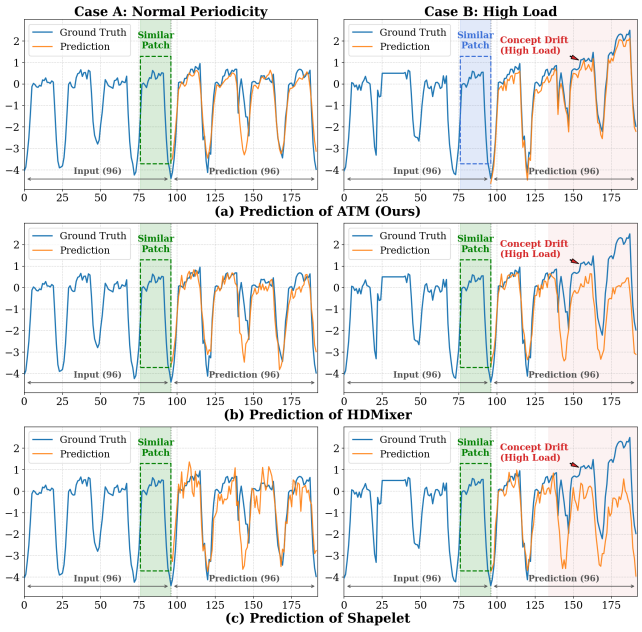



**Figure 1: Case study of semantic modeling on ETTh1. (a) ATM uses global queries to capture dynamic context, assigning distinct semantics (green vs. blue) to similar patches to predict the surge in Case B. (b, c) HDMixer and Shapelet assign identical semantics (same green color) due to traditional attention/shape matching, where numerically similar patches capture similar context, failing to adapt to context change.**

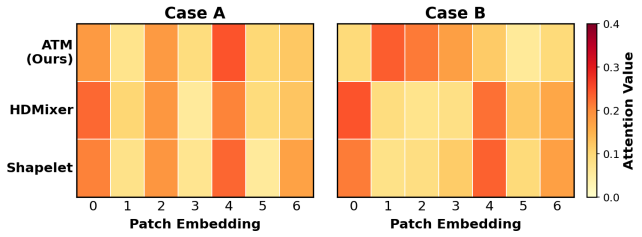



**Figure 2: Attention heatmaps of Case A and B. For numerically similar patches, ATM captures different contexts across cases through semantic modeling. In contrast, HDMixer and Shapelet focus on similar contexts in both cases.**

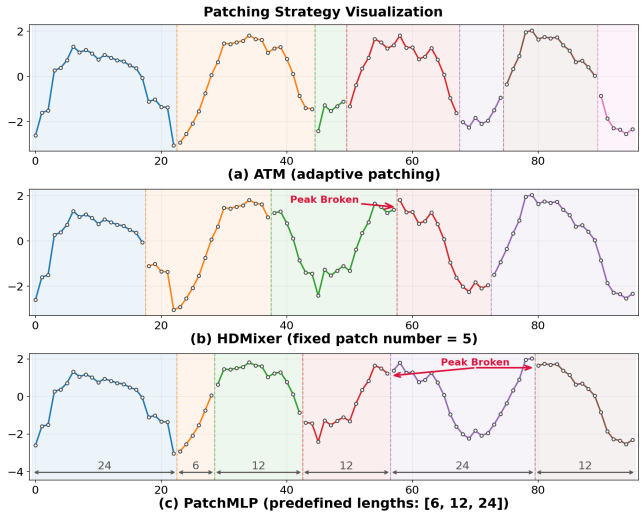



**Figure 3: Visualization of patch partitioning on ETTh1. ATM adaptively preserves complete temporal patterns (e.g., peaks), whereas HDMixer (fixed patch number) and PatchMLP (predefined lengths) destroy temporal integrity.**

**Table 1: Ablation on different modules on ETTh1, ETTm2 and Weather datasets. Replace-ATM Encoder denotes replacing it with a standard Transformer.**

| Design | Time Tokenizer | ATM Encoder | ETTh1 | | ETTm2 | | Weather | |
|---|---|---|---|---|---|---|---|---|
| | | | MSE | MAE | MSE | MAE | MSE | MAE |
| **ATM** | ✓ | ✓ | **0.426** | **0.422** | **0.277** | **0.318** | **0.249** | **0.266** |
| w/o | w/o | ✓ | 0.435 | 0.436 | 0.289 | 0.328 | 0.258 | 0.277 |
| | ✓ | replace | 0.429 | 0.427 | 0.281 | 0.321 | 0.252 | 0.270 |

**Table 2: Efficiency comparison of different models on ETTh1 with series length of 720.**

| Model | Parameters (M) | GPU Memory (GB) | Training Step Time (ms) | Inference Step Time (ms) |
|---|---|---|---|---|
| TimesNet | 1.71 | 1.5 | 132.40 | 30.34 |
| HDMixer | 1.05 | 1.1 | 80.86 | 24.79 |
| DualSG | 2.14 | 1.6 | 136.96 | 32.09 |
| **ATM** | 1.58 | 1.5 | 124.32 | 26.91 |

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**Table 3: Average performance comparison between ATM and HDMixer. The best performance is highlighted in red, and the second-best is underlined. Statistical significance is evaluated by paired t-test.**

| Models | | ATM | | HDMixer | | Significance |
|---|---|---|---|---|---|---|
| Metric | | MSE | MAE | MSE | MAE | p-value |
| ETTh1 | \|Avg | 0.426 ± 0.003 | **0.422 ± 0.002** | 0.435 ± 0.004 | 0.430 ± 0.003 | 0.021 |
| ETTh2 | \|Avg | 0.358 ± 0.002 | **0.384 ± 0.003** | 0.351 ± 0.002 | 0.385 ± 0.002 | 0.023 |
| ETTm1 | \|Avg | **0.379 ± 0.003** | **0.382 ± 0.003** | 0.390 ± 0.003 | 0.389 ± 0.004 | 0.019 |
| ETTm2 | \|Avg | **0.277 ± 0.002** | **0.318 ± 0.004** | 0.286 ± 0.003 | 0.327 ± 0.003 | 0.024 |
| Traffic | \|Avg | 0.442 ± 0.004 | **0.275 ± 0.002** | 0.424 ± 0.003 | 0.280 ± 0.002 | 0.045 |
| Weather | \|Avg | **0.249 ± 0.003** | **0.266 ± 0.003** | 0.257 ± 0.004 | 0.274 ± 0.004 | 0.030 |
| Electricity | \|Avg | 0.181 ± 0.002 | **0.265 ± 0.003** | 0.177 ± 0.002 | 0.272 ± 0.003 | 0.041 |